Trust, Measured: Verification-First AI Knowledge Extraction for Interpretive Corpora, Demonstrated on Jung's *Collected Works*

Damian Spendel *AI use disclosure — tools and roles: Claude Opus 4.8 (extraction), Claude Fable 5 (verification), Gemini 3.1 Pro (cross-vendor audit), orchestration via Claude Code, all under the author's direction. Detail in §6 and the Authorship Note.*

v1.0 — 27 July 2026 · DOI: 10.5281/zenodo.21631523 · Artifact: symbolicworld.observer · Repository: github.com/damianspendel/symbolic-world, release `v1.0.0`

Abstract

AI makes it cheap to turn twenty volumes of an author's writing into a queryable map of claims. Unguarded, it also makes it wrong: language models extract knowledge-graph facts at 30–66% accuracy and hallucinate 11–57% of citations in deployed systems. This paper contributes, primarily, a methodology for doing it trustworthily, and secondarily the artifact that demonstrates it: a knowledge graph of C. G. Jung's *Collected Works* (233 concepts and symbols, 634 relations, 659 references, each anchored to a numbered paragraph with a verbatim quote of at most 25 words, and each typed by voice: the author's own claim, doctrine he reports, or a source he quotes). The method treats extraction as risk management: seven named risks, a mitigation in the pipeline for each, and an experiment measuring each mitigation. No claim is published without passing a deterministic quote check, review by a model from a different family than the proposer, and audit by a model from a different vendor. Every published claim ships with its evidence and with live channels for any reader to dispute or confirm it. The compounded result: raw extraction got roughly one claim in four wrong; an independent vendor auditing a full published volume in its own words could uphold errors in one claim in a hundred, none of them fabrications, at about ten cents per verified claim. Run unchanged on a second, public-domain corpus (James's *The Varieties of Religious Experience*), the pipeline reproduces the same risk profile — structure caught at ceiling, voice the shared weak axis — and ships as a complete replication package. All error rates, corruption seeds, and adjudications are published with the artifact, along with the mechanisms a reader needs to check any claim against the book.

1. Opportunity

Every field has corpora like this: twenty volumes of an author arguing, quoting, and reporting, in which what the author *said*, on *whose authority*, at *exactly which place*, is the scholarly currency. Law has its judgments, philosophy its systems, history of science its debates. Our running example is C. G. Jung's *Collected Works*: twenty volumes of densely cross-referential prose in which the same symbol (Mercurius, the lion, salt, Luna) carries systematically different meanings by context and authority. The digital tools that exist are lexical: the ARAS concordance indexes word occurrences; Princeton's digital edition searches full text. But no resource answers the question a reader actually has: *what does Jung say the green lion is, and on whose authority?*

Large language models make the answer look easy, and cheap: an AI reads the volumes, drafts the claims, and a public atlas appears for about ten cents a claim (Figure 1).

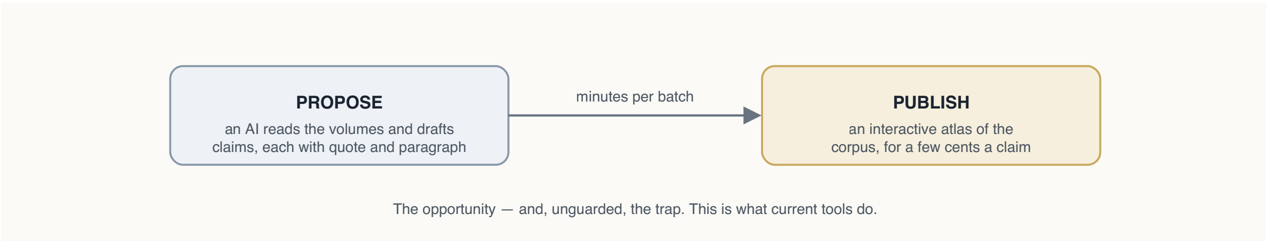

Figure 1 — The opportunity: let the machines do the hard yards. Unguarded, this is also exactly what current tools do; the rest of this paper is about why that is not good enough, and what to do instead.

The paper follows the shape of the work itself. The naive pipeline has named risks (§2), documented in the literature (§3). For every risk we inserted a mitigation into the pipeline (§4), and for every mitigation we ran an experiment to test whether it works (§5–6). §7 tables what risk remains, §8 what we haven't done, §9 the imperatives we draw. One design choice runs through everything: the trust chain is built to end in humans, not models, because a chain of models, however cross-checked, cannot certify itself. Honest status today: the machine part is complete and measured; the human part is live as channels, with accumulated human verification still small.

2. Risks

Reading is where the danger is. When a model turns prose into claims, seven things can go wrong, five in the reading itself and two in any machinery you build to catch them (Figure 2):

- **R1 — the statement is wrong or invented.** The model asserts something the text does not say.
- **R2 — the quote is fabricated.** The supporting quotation does not exist in the cited paragraph.
- **R3 — the meaning is wrong.** Right ingredients, wrong dish: direction reversed, wrong object, "is" where the text says "is like".
- **R4 — the context is wrong.** The claim is real but the *voice* is misassigned: the author's own view (*author-asserts*), doctrine the author reports (*author-reports*), or a source the author quotes (*author-quotes*), and how firmly it is said. (In Jung: is this his psychology, or the alchemists' doctrine he is describing?)
- **R5 — the selection is arbitrary.** Why these claims and not others? A different run might paint a different picture.
- **R6 — the checkers share blood with the writer.** If proposer and verifier come from one vendor, they may share blind spots that no one inside the loop can see.
- **R7 — the referee is one of ours.** When checkers disagree, some judge must rule, and if that judge is also our model, it is marking its own homework.

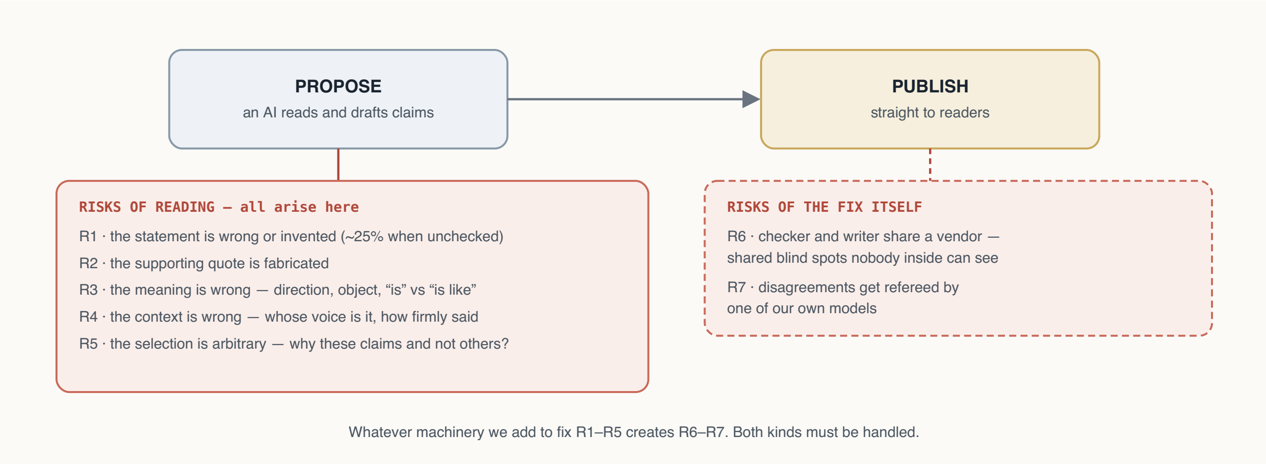

Figure 2 — The same naive pipeline, with its risks made visible. R1–R5 arise at the reading step; R6–R7 arise from the fix itself.

3. Related Work

The risks above are documented, not hypothetical:

LLM KG extraction. KGGen, GraphRAG, OpenIE establish unverified accuracy baselines (30–66%); schema-aware triple verification and prover–skeptic dialogue approaches add post-hoc verification. We differ in making verification a *mandatory publication gate*, grounding it in the full source paragraph, and calibrating the verifier itself with seeded corruptions.

Attributed generation. Verbatim-evidence systems (FullCite / structured inline citation, Quote-Tuning) enforce quotation at generation time; we enforce it at dataset level as a hard deterministic check, independent of any model.

LLM-as-judge reliability. Self-preference bias (measured −38% to +90%) and debiasing work motivate our hard constraint that proposer and verifier come from different model families; our seeded-corruption calibration (§6.2) is, to our knowledge, the first published sensitivity/specificity measurement of a verification gate in a humanities KG setting.

Industry practice. Production extraction pipelines converge on the same skeleton independently: schema-first design, typed mechanical validation with human escalation carrying the best-effort extraction (production document-pipeline lessons), continuous calibration of automated judges against sampled expert judgment (HITL evaluation practice), and verification loops with measured diminishing returns after ~2 rounds (verification-loop patterns). Dedicated KG fact-verification agents (AgentKGV) treat verification as a first-class subsystem. Our pipeline is the documented, measured instance of this consensus applied end-to-end to an interpretive humanities corpus: orthodox in its skeleton (cross-model judging, staged validation, audit artifacts) and novel in four places: publication-gating, seeded-corruption calibration of the judge itself, per-axis specialist decomposition, and rhetorical-voice provenance.

What we borrowed. The reference format adapts the assertion/provenance/publication-info anatomy of nanopublications (a native serialization is planned). Nanopublications have seen almost no humanities uptake[1]; this project is evidence that they deserve consideration as a publishing form for the humanities: the atomic unit of a nanopublication (one claim, its exact source, who stands behind it) matches what a humanities footnote has always tried to be, and the voice dimension interpretive corpora add fits naturally in its provenance graph. And the voice-typing treats provenance as *epistemic stance* in the sense of provenance-enhanced statements: our stubborn "sympathetic reportage" residual is precisely their claim-in-permeation between cognitive worlds (and an instance of Drucker's argument that humanities "data" is better understood as *capta* — taken, interpreted — than given). The nearest existing artifact is the Darshana Graph, whose author names a randomly sampled precision evaluation "the most valuable immediate extension" of that work; this paper operationalizes exactly that extension, with the caveat that our annotators are so far models from two vendors rather than multiple humans (human in the loop live as a channel, §8). To our knowledge, no knowledge graph of Jung's Collected Works previously existed[3]; the ARAS concordance and Princeton's digital edition are search, not relations.

4. Mitigants

Each risk in §2 forced a step into the pipeline. Figure 3 shows the result: the two-box dream of Figure 1 with four inserted checks, plus practices for the two risks that don't fit in a box.

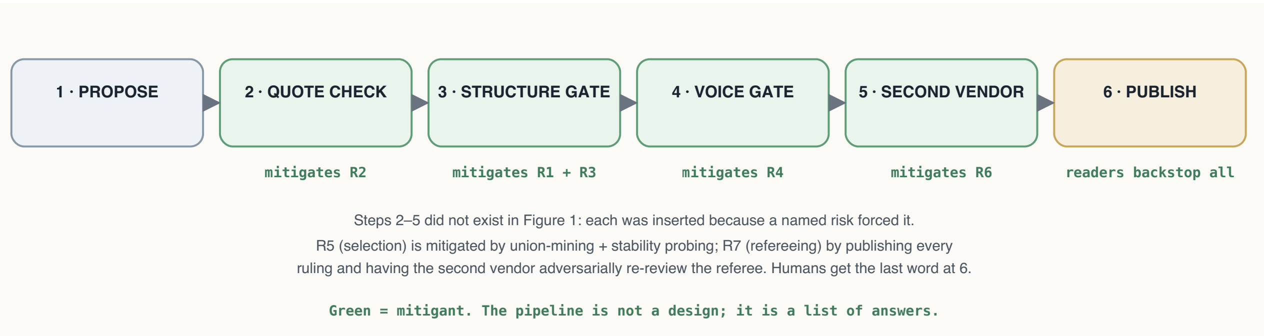

Figure 3 — Steps 2–5 did not exist in Figure 1; each was inserted because a named risk forced it (green tags name the risk each step mitigates).

Table 1 — the pipeline, step by step, and the risk each step mitigates.

Step	What it does	Mitigates	What comes out
1 · Propose	A proposer model reads a window of the corpus and drafts candidate claims, each with a verbatim quote and its anchor, typed *author-asserts / author-reports / author-quotes*	— (the risks arise here)	12–22 candidates per window
2 · Quote check	Deterministic: the quote must appear in the cited passage as a letter-normalized, in-order, near-contiguous token run (bounded gaps absorb interleaved footnote markers; collage quotes rejected, canary-guarded)	**R2**	candidates with real quotes
3 · Structure gate	A reviewer model **from a different family** reads the full passage: is the claim there, right way round, right object, no inflation of analogy into identity?	**R1 · R3**	SUPPORTED / PARTIAL + correction / WRONG
4 · Voice gate		**R4**	

Step	What it does	Mitigates	What comes out
	A second, narrow review: whose claim is it, and how firmly is it made?		voice + hedge corrections
5 · Second-vendor check	A model **from a different vendor** re-judges the batch, blind	**R6**	independent verdicts, disagreements adjudicated
6 · Publish	Corrections applied; WRONG dropped and logged; provenance stamped; integrity tests + canaries; build **fails closed** on any unverified reference; readers confirm/dispute	backstop for all	the public artifact

R5 (arbitrary selection) is mitigated by practice rather than a step (union-mining plus stability probing); R7 (self-refereeing) by publishing every ruling and the auditor's adversarial re-review of the referee.

Every batch's passage through these steps is committed to a **transaction ledger**, one commit per transition; the construction history is replayable and auditable commit-by-commit.

The cross-vendor loop. Beyond the per-batch check, the second-vendor auditor re-audits the published graph, and the machinery itself, **at every dataset release** (two full runs so far, E6 and E7; this is a release-gated commitment, not a background process): it judges samples blind (in two modes: under our rubric for comparability, or entirely in its own terms for independence), every disagreement is adjudicated against the source passage with the ruling logged, the auditor then **adversarially re-reviews the adjudicator's rulings** (the conflict-of-interest control), and finally it critiques our gate prompt itself for bias. Audit findings re-enter the same merge machinery as ordinary batch verdicts: there is no separate, weaker path into the graph.

5. Risk Measurement

Every mitigant gets measured, by a small set of reusable experiment designs, none of them corpus-specific:

- **Retro-verification.** Run the gates over output that was published *without* them: measures the raw risk the pipeline exists to remove.
- **Seeded-corruption calibration.** Deliberately break known-good entries (reverse a direction, swap an object, inflate an analogy, flip a voice), hide them among clean ones, and count what each gate catches: turns "we have a verifier" into "we have a verifier with a measured catch rate per error type."
- **Adversarial audit.** Instruct a reviewer to *break* each published claim: measures what survived the gates.
- **Independent re-extraction.** Re-run the proposer blind on an already-mined window: measures whether the selection is stable or arbitrary.
- **Cross-vendor audit, two modes.** A different vendor's model re-judges published samples, once under our rubric (comparable) and once entirely in its own terms (independent): measures shared blind spots, and whether the rubric itself steers verdicts.
- **Standing self-tests.** Corruption canaries the validators must catch on every run, and the human confirm/dispute channels on every published claim.

Figure 4 pins each design to the pipeline step it measures. Results for the Jung case study are consolidated in Table 3 (§10); per-experiment details are in Appendix A.

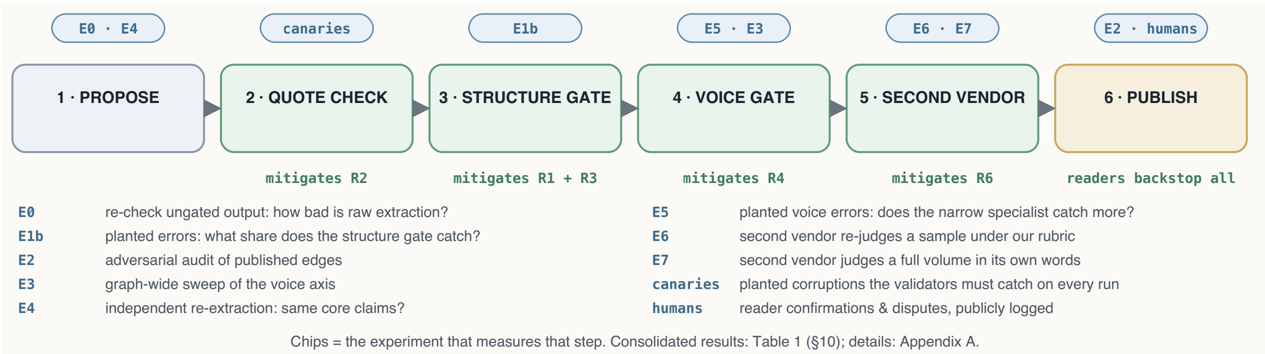

Figure 4 — Blue chips name the experiment that measures each step, with a one-line legend for each experiment. No box is taken on faith.

6. Case Studies

6.1 Jung's *Collected Works*

Corpus. Paragraph records `{volume, §, text, page}` extracted mechanically from epub markup (§ numbers read from bracketed markers, strictly ascending; never inferred), with a §→Bollingen-page concordance. The corpus stays local; only ≤25-word verified quotes are published (legal basis: `docs/LEGAL.md`).

Roles. Proposer: Claude Opus 4.8 (Anthropic). Structure and voice gates: Claude Fable 5 (Anthropic, a different family from the proposer). Second-vendor auditor: Gemini 3.1 Pro (Google). Orchestration: Claude Code, under the author's direction. The graph built before the cross-vendor stage existed was audited retroactively, in a stratified sample under the shared rubric (E6) and a full volume rubric-free (E7), which is what validated promoting that check from experiment to pipeline stage; the cross-vendor check itself costs only ~a tenth of a cent per citation[2] (the ~ten-cent figure quoted elsewhere is the full pipeline), so there is no economic reason to leave it post-hoc.

Schema. Nodes `{id, type ∈ {Concept, Operation, Symbol, Figure, Substance, Motif}, label}`; edges `{subject, relation (open vocabulary, verbatim-faithful), object, references[]}`; references `{volume, §, quote, claim_type, source?, confidence, verified, verified_by, verified_date}`.

Instantiation. The generic voice vocabulary becomes `jung-asserts` / `jung-reports-parallel` / `jung-quotes-source` (+ named source). Current state: **233 nodes · 634 edges · 659 references** across 468 distinct paragraphs of nine CW volumes; CW 14 covered end-to-end; 438 asserts / 160 reports / 61 quotes-source over 74 named sources; 84 hedged (medium-confidence) references; 100% gate-verified with per-reference verifier and date.

The artifact. The public Atlas (symbolicworld.observer) renders the graph in six typed regions with search and walkable citations. Every citation expands to its verification record (claim type, source, confidence, verifier, date, and a check-it-yourself pointer to the exact § and Bollingen page), so refutation needs only the printed edition and about two minutes. Two gated reader channels close the loop: **dispute** (a prefilled report to the public issue tracker; admissible disputes are re-judged with the objection attached; outcomes published in the verification log and linked from the edge) and **confirm** (reader confirmations enter the edge's public record as human-in-the-loop provenance). Accumulated upheld disputes above the published error bar would falsify the pipeline claim itself, not just individual edges.

What the measurements say about Jung. The residual error class maps a real property of the text: every model configuration, across two vendors, fails on the same passages, where Jung reports alchemical or Gnostic doctrine and half-adopts it. That blended voice is a finding about the *Collected Works*, not just about models (developed as Imperative 3, §9), and it marks the exact place where Jungian scholarship is needed.

6.2 Replication: James's *The Varieties of Religious Experience*

The identical pipeline, unchanged, was run on a second corpus: William James's *Varieties* (1902, public domain; 1,277 anchored paragraphs), mining the Conversion and Mysticism lectures. Results: 40 candidates; 40/40 quotes passed the mechanical check; the structure gate returned 37/3/0; the cross-vendor check passed all 38 published edges; and a seeded-corruption calibration (n=60) measured the gate at **90% sensitivity (CI 77–96%) and 100% specificity**, with the per-class profile matching Jung's: structural corruptions at ceiling, **voice again the weak axis (7/10, vs Jung's 18/25)**. The risk profile is not a Jung idiosyncrasy; for authored interpretive prose, voice appears to be the shared weak axis. Because the corpus is public domain, this replication ships complete: raw text, anchored corpus, candidates, verdicts, graph, and calibration (`james/` in the repository; visual graph at symbolicworld.observer/james.html). Full detail: `docs/experiments/exp8_james.md` (E8). Two honesty notes: the miner's instructions carry the claim-typing lessons of the Jung pipeline (its 7.5% raw error rate measures a taught miner, not a naive one), and this is pilot scale (38 edges, per-class calibration n=10).

7. Residual Risks

Table 2 answers one question: **of the risks in §2, what remains after the mitigations of §4, measured by the experiments of §5 (details: Appendix A)?** (A 16-item assumptions register is published in `docs/ASSUMPTIONS.md`.)

Table 2 — residual risks.

#	Risk	Mitigation in force	Evidence / data	Residual
i	**Shared-substrate risk** — proposer and verifier share one vendor; correlated blind spots invisible to both	Proposer and verifier from different families; a second-**vendor** check in the pipeline and at every release (E6–E7)	0 unsupported edges (n=100); 1.0% upheld residual (n=407); both vendors miss the *same* voice items → text ambiguity, not vendor blind spot	Low; one second vendor
ii	**Single translation** — quotes are Hull/Bollingen-bound	§ anchors are edition-stable; documented as assumption #1	—	German *GW* cross-check deferred (§8)

#	Risk	Mitigation in force	Evidence / data	Residual
iii	**Salience sampling** — "strongest relations" per window, not exhaustive parse	Stability probe; union-mining; claim-coverage framing (§6.6)	E4: independent re-extraction recovers core claims (37.5% strict pair overlap, theses stable); union batch: 13/20 re-findings	Tier-2 claims sampled, not complete
iv	**Calibration scope** — sensitivity measured only on four seeded corruption classes	Classes chosen from observed production errors; expanded to n=200	E1b: 86% (CI 78–91%) / 93% spec; per-class 72–96%	Unknown error types unmeasured, by construction
v	**Single-paragraph judging context** — cross-paragraph claims can be mis-scored	Documented; flagged cases adjudicated case-by-case	E7: 2 of 10 moderate flags were cross-paragraph context cases (both-defensible)	Standing; a windowed-context gate would cost ~2×
vi	**Adjudication discretion** — the orchestrator (an Anthropic model) rules on disputes	All rulings published verbatim; adversarial cross-review by the second vendor	E6: cross-review accepted 9/11 rulings and forced 2 stronger corrections	Cross-review is one round; full independence awaits the human tier
vii	**Shared-rubric convergence** — agreement partly an artifact of handing the auditor our rubric	Rubric-free replication mode (E7)	Identical detection profile with and without rubric; 95% cross-mode verdict consistency	Measured ≈ zero on tested classes
viii	**Human in the loop still thin** — channels live, accumulated human data ≈ zero	Confirm/dispute channels shipped; community pilot planned (§8)	Channels e2e-tested; 0 reader confirmations to date (stated plainly)	The main open gap — see §8

8. Limitations

Where §7 quantifies what remains of the *named* risks, this section lists the boundaries the experiments never touched.

Coverage beyond CW14/CW12 is thin (eleven volumes untouched) and the relation vocabulary remains open. Human-in-the-loop verification is live as channels but thin as data. Generality is demonstrated only at pilot scale (38 edges, one second corpus). Next, in order: the community verification pilot (`docs/ASSUMPTIONS.md` §E); a native nanopublication serialization; a German *Gesammelte Werke* cross-check.

9. Imperatives

1. **Ensure extraction is verified: the checking is the product.** A quarter of unverified extractions were flawed; after gating, adversarial audits find zero content errors.

2. **Know the two kinds of mistakes: machines only catch one well.** *Shape* mistakes (who does what to whom: wrong direction, wrong object) are caught almost every time (92–96% in calibration). *Voice* mistakes (who is speaking: Jung himself, or the alchemists he is describing) are caught only about three times in four. So nearly everything that slips through is a voice

mistake. Like proofreading: the spellchecker catches spelling nearly perfectly; whether a sentence is sincere or sarcastic survives it.

3. **Voice is genuinely hard, for machines and probably for people.** "Sympathetic reportage" (doctrine Jung reports *and* half-adopts) resists a clean three-way label; every model configuration we tested, across two vendors, missed the same borderline items.

4. **Teach the miner: fold the gate's corrections back into its instructions, not its weights.** When the verifier corrects a batch ("you typed reported doctrine as Jung's own claim"), those corrections are folded into the *next* batch's extraction instructions as explicit rules and examples. The extraction model never changes; its briefing does. Result: error flags fell from 18.9% on the oldest batches to 5.6% on the newest, and recent batches pass 20/20.

5. **Specialize the checkers: one measured weak axis each.** A checker given five things to check does the obvious ones well and the subtle one poorly; given one thing, it does that one well. On the same 25 planted voice errors, the five-criteria generalist caught 18 and the single-axis specialist 20 (paired discordance 3:1 in the specialist's favor, consistent in direction across five measurements including the other vendor's replay, modest in size, and not statistically resolved). The four errors neither ever catches are the sympathetic-reportage boundary: a property of the text, not the prompt.

6. **Measure coverage in claims, not pages: the graph is accurate, not exhaustive.** It cites 34% of CW 14's paragraphs, yet independent re-extraction recovers each chapter's core claims (E4), because most paragraphs argue, illustrate, or amplify rather than assert a new relation. Like a topographic map, it is complete above a chosen prominence, not in every hill; union-mining lowers that threshold at ~$2 per window.

7. **Make trust inspectable.** Per-citation verification records, published error rates, self-testing validators, and live dispute/confirmation channels convert "trust us" into "check us."

10. Summary of Results

The compounded result, in one sentence: raw AI extraction on this corpus gets roughly one claim in four wrong; after the pipeline, an independent vendor auditing a full volume in its own words could uphold errors in one claim in a hundred, a ~25-fold error reduction, with every surviving error a nuance of attribution rather than a fabrication or reversal, at about ten cents per verified claim, and every claim checkable by any reader in about two minutes.

Table 3 — pipeline step · risk · mitigant · experiment · headline result (Jung), with the James replication (E8) alongside.

Pipeline step	Risk	Mitigant	Experiment	Result (Jung, CW)	Result (James, VRE)
1 · Propose	R1 wrong/ invented statements	never publish raw — everything below	E0	unchecked extraction: ~25% flawed	taught miner: 7.5% raw (not comparable — see E8)
1 · Propose			E4		— (not run)

Pipeline step	Risk	Mitigant	Experiment	Result (Jung, CW)	Result (James, VRE)
	R5 arbitrary selection	union-mining; claim-coverage framing		independent re-extraction finds the same core claims	
2 · Quote check	R2 fabricated quotes	deterministic verbatim match	canaries	fabrication eliminated; canaries caught on every run	40/40 quotes verbatim
3 · Structure gate	R3 wrong meaning	independent model family reads the full paragraph	E1b	86% caught (CI 78–91%); 92–96% on structure	90% caught (CI 77–96%); structure 29/30
4 · Voice gate	R4 wrong context	narrow specialist on the measured weak axis	E5 · E3	sweep fixed 10.3%; specialist 80% vs 72% paired	voice-flips 7/10 — same weak axis
5 · Second vendor	R6 shared blind spots	Gemini re-judges — with our rubric and without	E6 · E7	0 unsupported /100; 1.0% residual /407; rubric effect ≈ 0	38/38 supported (pre-merge)
(refereeing)	R7 marking own homework	rulings published verbatim; second vendor re-reviews the referee	in E6	9/11 rulings upheld; 2 stronger corrections forced	— (no disputes arose)
6 · Publish	whatever still got through	adversarial audit; evidence views; reader confirm/dispute	E2 · humans	0 content errors in 30; human data early	published graph + full reproduction package

A second-corpus column is planned: the same harness run on a public-domain corpus (William James, ***The Varieties of Religious Experience****), yielding a per-corpus risk profile alongside Jung's, and a fully distributable reproduction package.*

11. Conclusion

What has this methodology actually produced? Judged dimension by dimension:

Accurate, by independent measurement: when a second vendor audited a full volume in its own words, it could uphold errors in about one claim in a hundred, none of them fabricated; the unguarded pipeline the field currently runs gets roughly one in four wrong. Complete only in a defined and tested sense. The core claims of covered chapters, the ones independent re-extraction re-finds, are there; the long tail below that line is sampled, deliberately. It is not comprehensive: one volume end-to-end, eight more touched, eleven untouched. It is inspectable, always, since every claim carries its evidence, verifier, and history, the construction replays commit by commit, and disputing takes two minutes and a book. General, now demonstrated at pilot scale, because the identical pipeline ran unchanged on a second, public-domain corpus (James's *Varieties*), reproduced the same risk profile, and ships as the complete reproduction package the Jung corpus cannot legally provide. And human-verified only thinly so far. The channels exist and are tested; the accumulated human record is the project's youngest and most important open front.

None of these dimensions is the point on its own. The point is that each now comes with a number or a mechanism where an adjective used to be: accuracy has an error bar, completeness has a tested definition, inspection has a button, and the gaps are stated in the artifact itself. The machines did the hard yards (proposing, checking, cross-checking, and the bookkeeping of every correction) at about ten cents per verified claim[2]. Human judgement has the last word, and, for the first time on this corpus, the tools to exercise it.

The method is not tied to Jung. It should carry to any corpus where the scholarly work is establishing who said what, and exactly where: law, philosophy, theology, the history of science. A first replication on James's *Varieties* (E8) supports this at pilot scale, and — being public domain — ships as the complete reproduction package the Jung corpus cannot legally provide. Scaling that replication is the next test.

Authorship note

Conception, direction, corpus provision, publication decisions, and final responsibility: **Damian Spendel**. Edge proposal: *Claude Opus 4.8* (Anthropic). Independent verification, audits, calibration runs: *Claude Fable 5* (Anthropic). Cross-vendor auditing, adjudication cross-review, and prompt-bias critique: *Gemini 3.1 Pro* (Google). Orchestration, tooling, experiments, drafting: *Claude Code* (Anthropic), under the author's instruction. Proposer/verifier family separation and the second-vendor audit tier are deliberate design constraints.

Artifact availability. Dataset, pipeline code, transaction ledger, all experiment inputs/verdicts/adjudications, and this paper: github.com/damianspendel/symbolic-world (public; the issue tracker is the standing dispute channel described in §4). Live atlas: symbolicworld.observer.

Key references

- KGGen (NeurIPS 2025) · RAG vs GraphRAG
- Attribution, Citation, and Quotation: A Survey · Cited but Not Verified
- Self-Preference Bias in LLM-as-a-Judge · UDA
- Structured Inline Citation Generation · Schema-Aware Triple Verification · Prover–Skeptic KB Generation
- Nanopublications · Provenance-Enhanced Statements
- KGs in Libraries & Digital Humanities · ARAS Concordance

Appendix A — Experiment details

Risk 1 — raw AI extraction can't be trusted. *Mitigation: never publish raw extraction. Test: measure how bad it actually is.*

E0 — Retro-verification of ungated extraction (n=238). Everything mined before the gate became mandatory was re-checked by the standard gate: 178 confirmed, 54 corrected, 6 edges deleted. That is a **25.2% flaw rate for unverified LLM extraction** on this corpus, consistent with public benchmarks[4]. Dominant classes: voice conflation, identity overreach, referent drift, direction reversal; deletions included a spliced quote and a reversed symbolization.

Risk 2 — the gate itself might not work. *Mitigation: treat the verifier as an instrument and calibrate it; feed it deliberately broken entries, hidden among clean ones, and count what it catches. (The mechanical validators get the same treatment: five planted corruptions that must be caught on every test run.)*

E1 — Verifier calibration by seeded corruption (n=40, seed 20260726, blind). 20 intact controls + 20 corruptions (5 each: direction reversal, voice flip, object swap, identity overreach), quotes/paragraphs untouched so only the semantic gate is tested; production prompt; key withheld. Results: **specificity 20/20 (100%)**; sensitivity 15/20 (75%): **reversal 5/5, object swap 5/5**, overreach 3/5, **voice flip 2/5**. Missed voice flips were sympathetic-reportage cases (doctrine Jung partially adopts): genuinely ambiguous voice.

E1b — Expanded calibration (n=200, seed 20260727, blind). The per-class sample was expanded tenfold (25 corruptions per class + 100 intact controls; reversal corruptions restricted to directional relations: a design fix, since reversing a symmetric relation is not a corruption). Results with 95% Wilson intervals: **sensitivity 86/100 (86%, CI 78–91%)**, **specificity 93/100 (93%, CI 86–97%)**; reversal **23/25 (92%)**, object-swap **24/25 (96%)**, overreach 21/25 (84%), voice-flip **18/25 (72%, CI 52–86%)**. Two honest consequences: (i) E1's n=5 voice-flip estimate (2/5) was noisy: the generalist's true voice baseline is materially higher, which **narrows the measured generalist–specialist gap** (72% vs 5/6; directionally consistent across all four first-party and cross-vendor measurements, but not statistically resolved at current specialist n; the E5 claim is downgraded accordingly); (ii) measured specificity is a **lower bound**: the 7 flagged controls are production references, and several flags are plausible nuance findings of the familiar vocabulary/voice class; one was upheld and corrected in the graph. Full tables: `docs/experiments/exp1b_expanded_calibration.md`.

Risk 3 — the finished graph might still contain errors. *Mitigation: audit the published product in three independent ways: an auditor told to break each edge (E2), a graph-wide sweep of the weakest axis (E3), and an independent re-extraction to test whether the selection is arbitrary (E4).*

E2 — Adversarial audit of published references (n=30, seed 20260725). A differently-prompted reviewer instructed to *break* each edge: **0 content/direction/fabrication errors**; 4 disputes (13.3%), all attribution/modality refinements; all applied.

E3 — Full-graph modality sweep (n=634, complete). A narrow auditor checking only claim-type and hedge fidelity flagged **65/634 (10.3%)**: 50 claim-type corrections (including **5 in the reverse direction**: Jung's own theses mistyped as reportage, showing the audit is not a systematic deflation of the author's voice), 10 source corrections, 15 hedge downgrades, all applied. Flag rate by extraction age: **18.9% on the oldest edges → 5.6% on the newest**, and the three most recent production batches passed the gate **60/60**, evidence that gate corrections, fed back into extractor instructions, compound into first-pass precision. Three independent methods now converge on the modality error rate (~10–13%: audit 13.3%, calibration voice-flip deficit, sweep 10.3%).

E4 — Extraction stability probe (n=40 across two windows, blind). Two already-mined windows re-mined independently with a fixed "20 strongest relations" budget (shared node vocabulary; prior triples withheld): strict unordered {subject, object} pair overlap with the graph was **7/20 (35%)** on CW14 §371–440 and **8/20 (40%)** on CW12 §332–420 (**37.5% combined**). Core theses recur near-verbatim across runs; non-overlap decomposes into complementary genuine edges (the windows hold 29–33 graph edges each, more than one 20-edge budget can cover), schema-shape variants of the same insight, and one case where the probe independently re-found an edge that adjudication had dropped on schema grounds. The graph is therefore a *stable curated map*, not an exhaustive parse; union-mining is the costed coverage upgrade.

Risk 4 — the gate is weakest on one axis (voice). *Mitigation: add a second, narrow reviewer that checks only whose claim it is and how firmly it is made; then calibrate that too.*

E5 — Voice-specialist calibration (n=20, seed 20260727, blind). Per-axis seeded corruption of post-sweep ground truth (6 voice flips, 4 hedge strips, 10 clean controls) run through the narrow stage-2 auditor: **sensitivity 80%** (voice flips **5/6 = 83%**, hedge strips 3/4), **specificity 10/10 (100%)**. Against the generalist's 40% voice-flip detection on the same error class (E1), this is a direct, controlled measurement of what we shorthand *semantic diffusion* in verification. Task decomposition is well documented for generation (least-to-most prompting, chain-of-thought decomposition, plan-and-solve); our contribution is not the decomposition idea but its seeded-corruption *measurement on the verification side*, replicated cross-vendor: same model, same paragraphs, scope narrowed from five criteria to one; detection improved on paired items (2/5→5/6 first-party; 2/6→4/6 cross-vendor) at zero false-positive cost. The expanded calibration (E1b) revises the generalist voice baseline upward to 72%, narrowing the measured gap; we therefore state the specialist advantage as **directionally consistent across four measurements but pending a larger specialist calibration for statistical resolution**, an honest downgrade the larger sample forced, and an example of the methodology auditing its own earlier claims. The two residual misses are the same sympathetic-reportage boundary cases identified by E1 and E3. **E5b (paired expansion, n=25+25):** on E1b's 25 voice-flips, specialist 20/25 (80%, CI 61–91%) vs generalist 18/25 (72%); paired discordance 3:1, direction consistent across five measurements, effect modest, not statistically resolved; specialist specificity 24/25. Details: `docs/experiments/exp5b_specialist_expanded.md`.

Risk 5 — proposer and verifier come from one vendor, and could share blind spots no one inside can see. *Mitigation: bring in a different vendor's model as auditor, first with our grading scale, then (because the scale itself could bias it) with no scale at all.*

E6 — Cross-vendor verification audit (n=100 production edges + calibration replays, Gemini 3.1 Pro). A verifier from a different vendor (Google Gemini 3.1 Pro, temperature 0, same production gate prompt, blind to prior verdicts) audited a stratified sample of 100 published edges (strata: volume × claim type × extraction age): **89 SUPPORTED, 11 PARTIAL, 0 WRONG**; zero fabrications, reversals, or unsupported edges found by an independent vendor, and every disagreement confined to the attribution/referent-nuance axis the pipeline already identifies as its residual weakness. Adjudication of the 11 disagreements against the ground-truth paragraphs: **4 upheld** (two referent-precision errors, an edge anchored to a paragraph naming the *filius regius* rather than Rex and one naming Adam rather than Anthropos; one subject overreach, one referent demotion; all corrected, two near-duplicate edges removed), **5 both-defensible** (the sympathetic-reportage boundary), **2 not upheld** (one objection to ellipsis splicing permitted by the stated quotation rule, one strict-vocabulary reading). Because all sampled edges were published (the first-party verifier's marginal is degenerate), κ is uninformative here; the two-sided comparison comes from calibration replays: on the E1 corruption set Gemini scores **70% sensitivity / 95% specificity** (Fable: 75%/100%), with reversals and object swaps 5/5 and voice flips 2/5, **missing the same individual voice-flip items as Fable**, evidence that these items are intrinsically ambiguous rather than a vendor blind spot; on the E5 set the narrow voice brief lifts Gemini's voice-flip detection from 33% to 67% (Fable: 40% → 83%): **the semantic-diffusion effect replicates across vendors**. The orchestrator's dispute rulings were then adversarially reviewed by the cross-vendor model (9 AGREE / 2 PARTIALLY-AGREE / 0 DISAGREE); both partial agreements argued for stronger remedies than confidence demotion, and both were accepted (two referent retargets). Post-audit graph at the time of E6: 228 nodes, 608 edges, 633 references (the graph has since grown under the same pipeline rules; §4 states current counts). E6's headline is stated precisely: zero unsupported edges *under the shared rubric* (see residual risk vii).

E7 — Rubric-free cross-vendor audit (n=407, full CW14 + calibration replay, Gemini 3.1 Pro). E6's prompt-bias audit raised *shared-rubric convergence*: agreement measured under the first vendor's rubric partly reflects shared thresholds. E7 removes the rubric: no verdict enum, no claim-type definitions, no strictness persona; the second vendor judges every CW 14 reference (the full volume, n=407) in its own terms. Results: **356 supported / 47 partly / 4 no**; on the auditor's own severity scale, 350 none / 47 minor / 8 moderate / 2 serious. Adjudication of the 10 moderate+serious findings upheld **4 (1.0%)**: one misanchored edge deleted, one relation and one subject retargeted, one relation relabeled; none was a fabrication or reversal. The decisive measurement is the calibration replay: Gemini's rubric-free detection profile is **identical** to its with-rubric profile (70% sensitivity, 95% specificity, same per-class breakdown, same three missed sympathetic-reportage voice flips), and on the 58 doubly-audited references, verdict consistency across modes is **55/58 (95%)** with all three divergences in the *stricter* direction. Shared-rubric convergence, a legitimate concern, has a measured impact of ≈ zero on these error classes: detection is driven by the paragraph evidence, and the rubric's real function is output comparability, not verdict steering (residual risk vii, discharged with measurement).

Risk 6 — the referee of all these disagreements is itself a model from the first vendor. *Mitigation: publish every ruling verbatim, and have the second vendor adversarially re-review the referee's rulings (it accepted 9 of 11 and successfully forced 2 stronger corrections; see E6). Full independence arrives with the human tier (§8).*

Cost. From production telemetry (~97K tokens/miner batch, ~56K/gate batch at \$5/\$25 and \$10/\$50 per MTok): ≈ **\$0.09 per verified citation**, ≈ \$0.12 including retro-verification, audit, and calibration overhead; total artifact cost ≈ \$60–90.

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1. The ~10.8M nanopublications catalogued by Kuhn et al. (2018) derive from DrugBank, GloBI, DisGeNET, WikiPathways, neXtProt, OpenBEL, LIDDI, and the Human Protein Atlas — all life-science datasets; we are aware of no published humanities nanopublication dataset. ↩

2. Production telemetry: a mining batch consumes ≈83–97K tokens (proposer at \$5/\$25 per M tokens in/out), the structure gate ≈50–56K and the voice specialist ≈37K (verifier at \$10/\$50), the cross-vendor check ≈3K (≈\$0.01/batch); ≈20 candidates per batch yields

≈\$0.09–0.10 per verified citation at first pass, ≈\$0.12–0.14 with calibrations and audits amortized. Total spend for the artifact plus its complete evaluation suite (E0–E8, code and paper audits): ≈\$85–115. These figures are marginal API cost only; they exclude the author's and orchestrator's time (direction, adjudication, writing — roughly a person-week for this artifact). ↩↩

3. Based on searches (July 2026) of arXiv, digital-humanities venues, and Jungian digital resources; the nearest artifacts found are discussed above. ↩

4. The 30–66% fact-accuracy range for LLM triple extraction and 11–57% citation-hallucination rates cited in §3. ↩